# Drakens Energy 360

An end-to-end synthetic data engineering platform for a downstream energy business

Report generated 2026-09-08 · Warehouse: drakens360_test_full.duckdb

**Independent synthetic portfolio project.** Drakens Energy is a fictional company invented for this project. It does not exist. Every site, customer, employee, supplier, asset, coordinate, price, volume, margin and incident in this report is randomly generated. Site coordinates are South African town centroids plus random jitter and do not represent any real service station.

## Summary

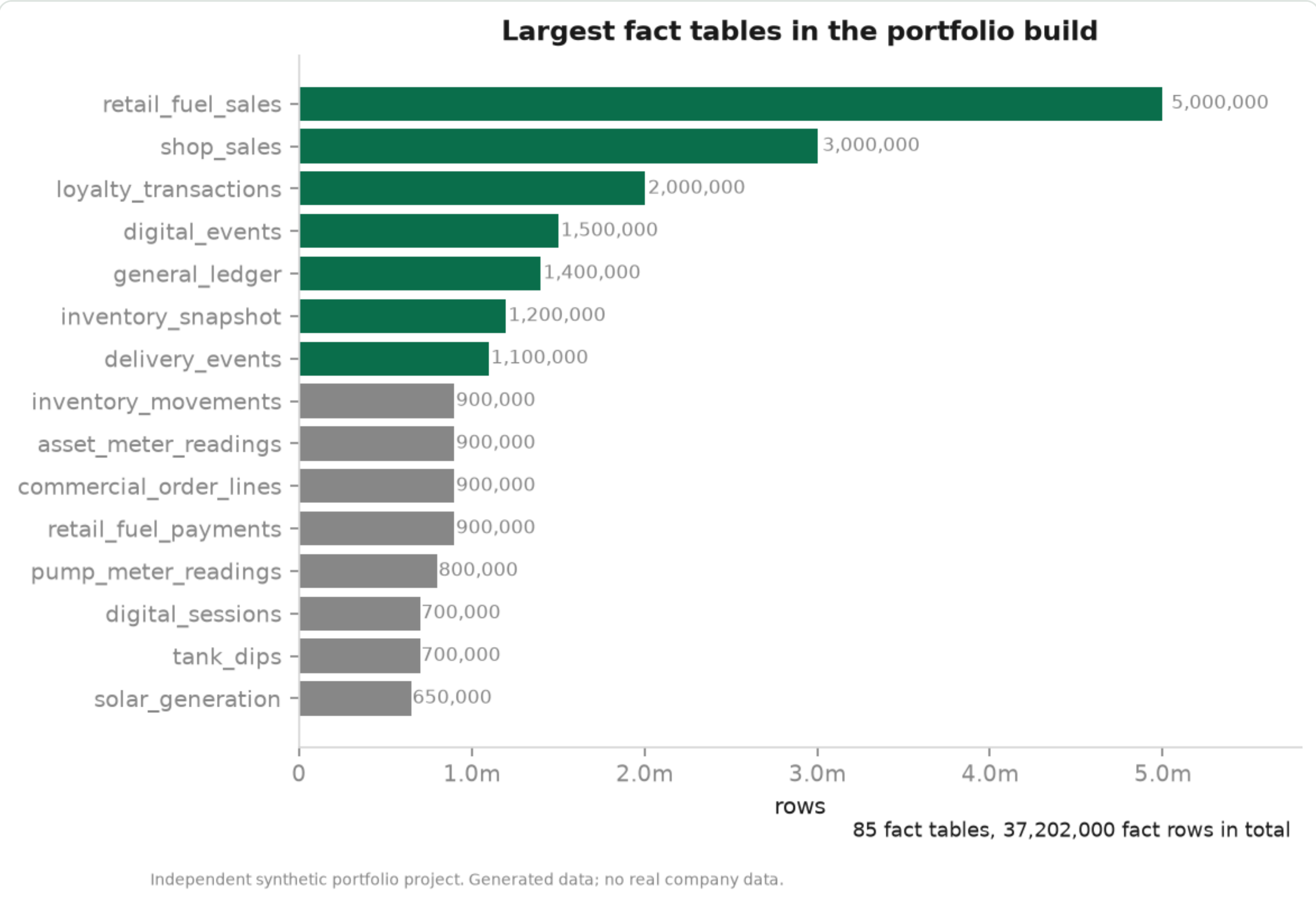
| FACT TABLES | FACT ROWS | DEFECTS INJECTED | PROVINCES |
|---|---|---|---|
| **85** | **37,202,000** | **62,924,526** | **9** |

The platform covers retail forecourt, commercial B2B, supply and distribution, LPG, lubricants, aviation, marine, loyalty, digital, EV and solar, asset maintenance, HSSEQ and finance — a downstream network of 587 synthetic service stations across 26 African markets, of which 250 are in South Africa spanning all nine provinces. It is built to demonstrate the parts of data engineering that are hard, not the parts that demo well: a deliberately dirty landing zone, a cleansing layer whose output is measured rather than asserted, and models reported against baselines they have to beat.

## Build

| Table | Rows |
|---|---:|
| fact_retail_fuel_sales | 5,000,000 |
| fact_shop_sales | 3,000,000 |
| fact_loyalty_transactions | 2,000,000 |
| fact_digital_events | 1,500,000 |
| fact_general_ledger | 1,400,000 |
| fact_inventory_snapshot | 1,200,000 |
| fact_delivery_events | 1,100,000 |
| fact_inventory_movements | 900,000 |
| fact_asset_meter_readings | 900,000 |
| fact_commercial_order_lines | 900,000 |
| fact_retail_fuel_payments | 900,000 |
| fact_pump_meter_readings | 800,000 |
| fact_digital_sessions | 700,000 |
| fact_tank_dips | 700,000 |
| fact_solar_generation | 650,000 |
| fact_customer_invoices | 620,000 |
| fact_site_energy_consumption | 600,000 |
| fact_customer_receipts | 560,000 |
| fact_commercial_orders | 550,000 |
| fact_mobile_payments | 520,000 |
| fact_site_daily_operations | 500,000 |
| fact_loyalty_points_balance | 480,000 |

| Table | Rows |
|---|---|
| fact_finance_revenue_cost | 420,000 |
| fact_forecourt_shift | 420,000 |
| fact_lpg_sales | 400,000 |

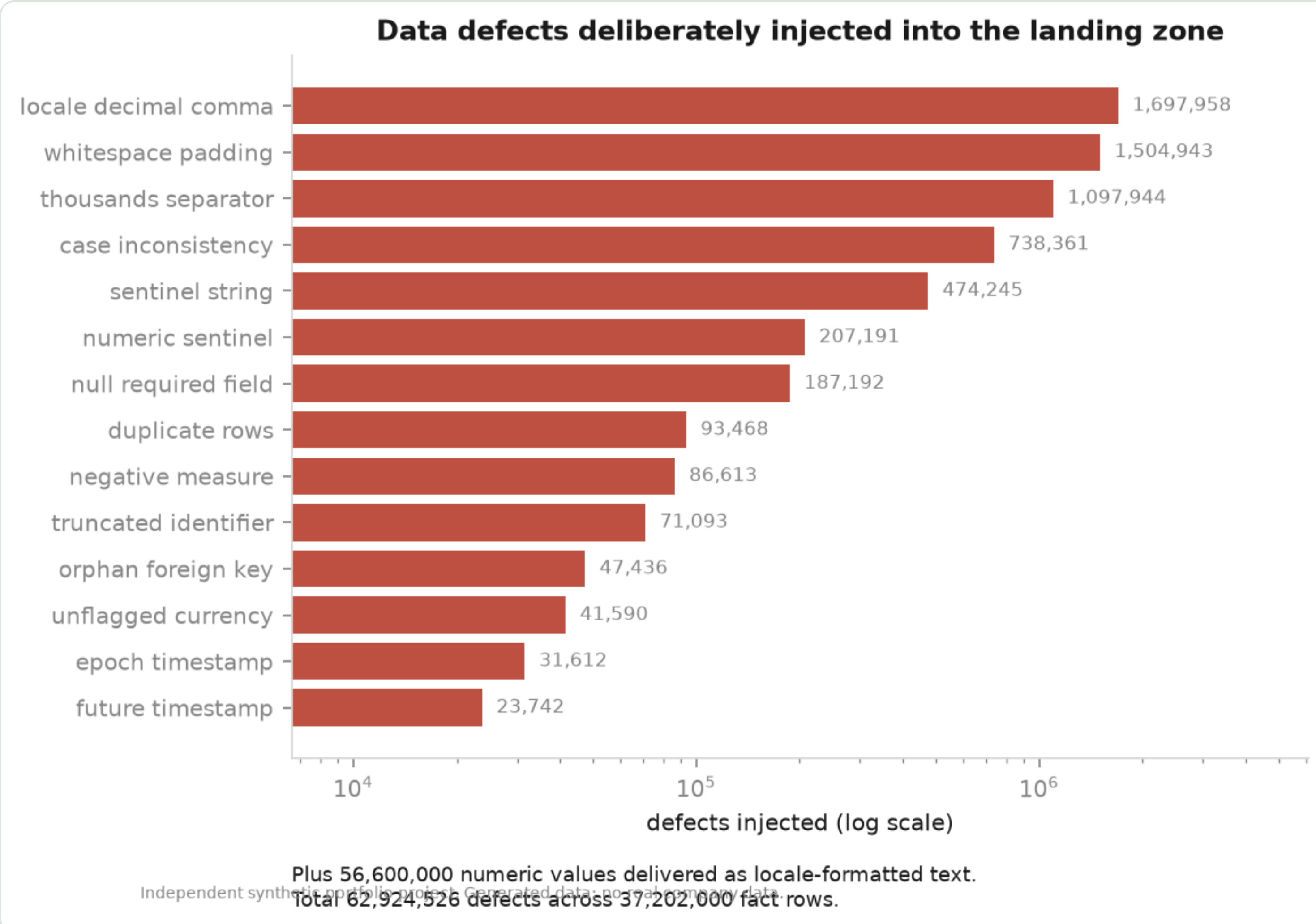



Largest fact tables in the portfolio build.

## Data quality

The landing zone is dirty on purpose. A portfolio project built on perfect data demonstrates nothing, because the difficult part of the job is deciding what to do with the row where litres reads "1 234,5", the site code has a trailing space, and the same transaction arrived twice because the point-of-sale replayed its batch. Every defect is counted as it is injected, which turns cleansing from an assertion into a measurement.

| Defect type | Defects |
|---|---|
| numeric delivered as text | 56,600,000 |
| locale decimal comma | 1,697,958 |
| whitespace padding | 1,504,943 |
| thousands separator | 1,097,944 |
| case inconsistency | 738,361 |
| sentinel string | 474,245 |
| numeric sentinel | 207,191 |
| null required field | 187,192 |
| duplicate rows | 93,468 |
| negative measure | 86,613 |
| truncated identifier | 71,093 |
| orphan foreign key | 47,436 |
| unflagged currency | 41,590 |
| epoch timestamp | 31,612 |
| future timestamp | 23,742 |
| derived value mismatch | 12,256 |
| unit of measure error | 7,233 |
| encoding mojibake | 1,649 |




Defects deliberately injected into the landing zone.

## Cleansing outcome

| Table | Raw rows | Cleansed | Quarantined | Duplicates removed |
|---|---|---|---|---|
| fact_retail_fuel_sales | 200,000 | 194,933 | 4,572 | 495 |
| fact_shop_sales | 120,000 | 116,702 | 2,992 | 306 |
| fact_general_ledger | 84,000 | 83,256 | 540 | 204 |
| fact_loyalty_transactions | 80,000 | 78,944 | 857 | 199 |
| fact_delivery_events | 66,000 | 65,557 | 283 | 160 |
| fact_digital_events | 60,000 | 59,593 | 236 | 171 |
| fact_retail_fuel_payments | 54,000 | 53,382 | 489 | 129 |
| fact_commercial_order_lines | 54,000 | 53,145 | 728 | 127 |
| fact_asset_meter_readings | 54,000 | 53,453 | 397 | 150 |
| fact_inventory_movements | 54,000 | 53,201 | 656 | 143 |
| fact_inventory_snapshot | 50,000 | 49,300 | 583 | 117 |
| fact_pump_meter_readings | 48,000 | 47,383 | 502 | 115 |
| fact_tank_dips | 42,000 | 41,351 | 543 | 106 |
| fact_digital_sessions | 42,000 | 41,745 | 162 | 93 |
| fact_customer_invoices | 37,200 | 36,733 | 374 | 93 |
| fact_site_energy_consumption | 36,000 | 35,623 | 305 | 72 |
| fact_customer_receipts | 33,600 | 33,205 | 301 | 94 |
| fact_mobile_payments | 31,200 | 30,842 | 284 | 74 |
| fact_site_daily_operations | 30,000 | 29,556 | 365 | 79 |
| dim_loyalty_member | 30,000 | 30,000 | 0 | 0 |
| fact_loyalty_points_balance | 28,800 | 28,491 | 232 | 77 |
| fact_solar_generation | 26,000 | 25,704 | 228 | 68 |
| fact_finance_revenue_cost | 25,200 | 22,476 | 2,662 | 62 |
| fact_forecourt_shift | 25,200 | 24,915 | 217 | 68 |

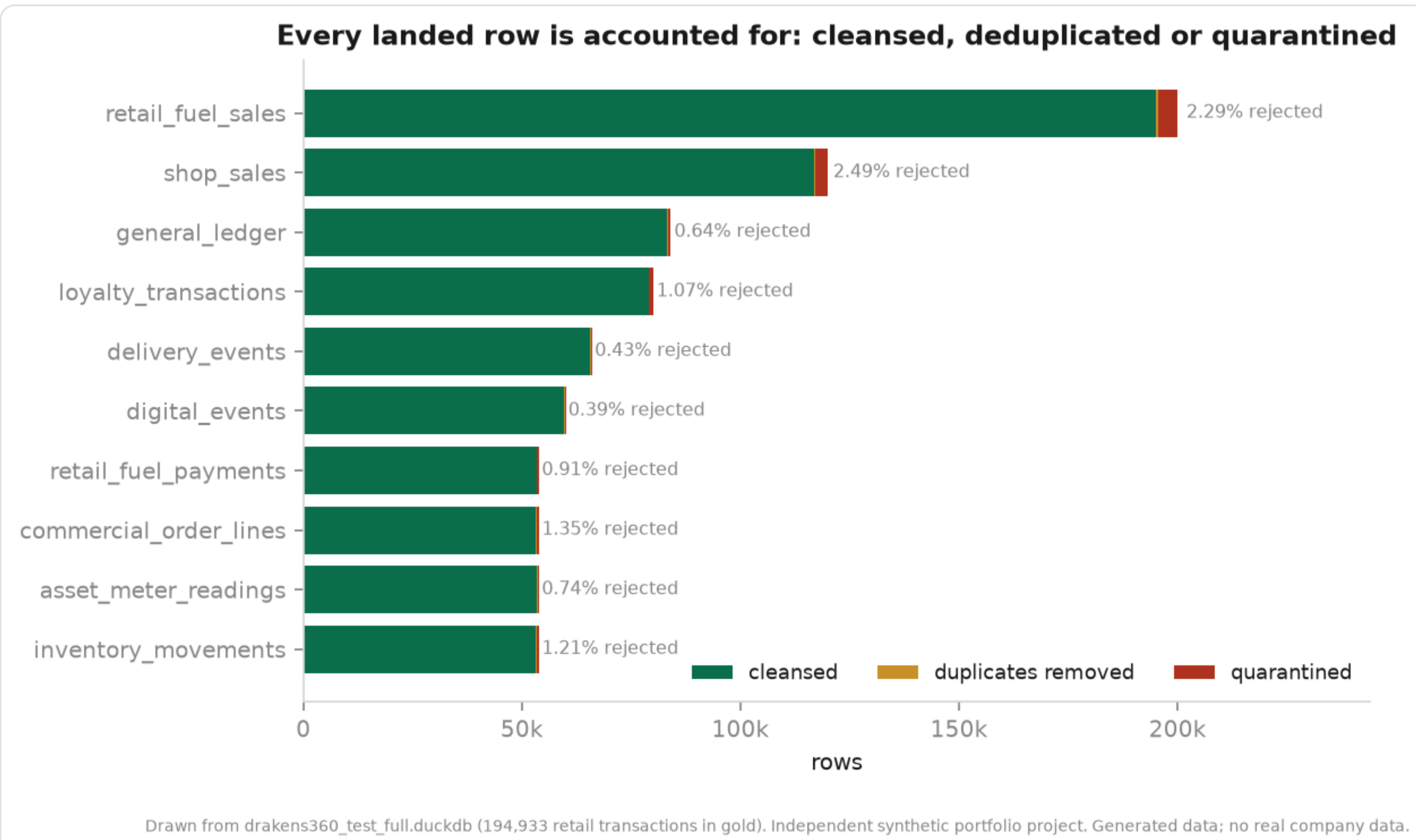
| Table | Raw rows | Cleansed | Quarantined | Duplicates removed |
|---|---|---|---|---|
| dim_digital_user | 25,000 | 25,000 | 0 | 0 |
| fact_commercial_orders | 25,000 | 24,417 | 528 | 55 |
| fact_ev_charger_status | 24,000 | 23,737 | 199 | 64 |
| fact_grid_import_export | 22,800 | 22,586 | 152 | 62 |
| fact_commercial_deliveries | 22,800 | 22,447 | 293 | 60 |
| fact_accounts_receivable | 22,800 | 22,508 | 232 | 60 |
| fact_daily_executive_kpi | 22,800 | 22,385 | 357 | 58 |
| fact_vehicle_trips | 20,400 | 20,247 | 100 | 53 |
| fact_opex | 19,200 | 18,995 | 164 | 41 |
| fact_customer_credit_exposure | 18,000 | 17,735 | 234 | 31 |
| fact_accounts_payable | 18,000 | 17,848 | 102 | 50 |
| fact_forecast | 18,000 | 17,739 | 222 | 39 |
| fact_lpg_cylinder_movements | 18,000 | 17,814 | 143 | 43 |
| fact_lpg_sales | 16,000 | 15,627 | 331 | 42 |
| fact_price_changes | 15,600 | 15,353 | 204 | 43 |
| fact_budget | 15,600 | 15,479 | 87 | 34 |
| fact_replenishment_orders | 15,600 | 15,362 | 192 | 46 |
| fact_campaign_responses | 15,600 | 15,470 | 92 | 38 |
| fact_training_compliance | 15,600 | 15,442 | 123 | 35 |
| fact_deliveries | 15,000 | 14,796 | 175 | 29 |
| fact_budget_vs_actual | 14,400 | 14,262 | 102 | 36 |
| fact_lubricant_sales | 14,000 | 13,687 | 281 | 32 |
| fact_terminal_issues | 12,600 | 12,463 | 95 | 42 |
| fact_asset_inspections | 12,000 | 11,879 | 93 | 28 |

| Table | Raw rows | Cleansed | Quarantined | Duplicates removed |
|---|---|---|---|---|
| fact_spare_parts_usage | 11,400 | 11,292 | 83 | 25 |
| fact_procurement_orders | 11,400 | 11,268 | 110 | 22 |
| fact_safety_observations | 10,800 | 10,715 | 55 | 30 |
| fact_fleet_costs | 10,800 | 10,704 | 59 | 37 |
| fact_terminal_throughput | 10,800 | 10,686 | 93 | 21 |
| fact_procurement_receipts | 10,000 | 9,858 | 109 | 33 |
| fact_energy_costs | 9,600 | 9,470 | 111 | 19 |
| fact_lubricant_orders | 9,000 | 8,882 | 94 | 24 |
| fact_route_performance | 9,000 | 8,926 | 56 | 18 |
| fact_permit_to_work | 9,000 | 8,910 | 73 | 17 |
| bridge_customer_contract | 8,947 | 6,000 | 0 | 2,947 |
| fact_terminal_receipts | 8,400 | 8,314 | 64 | 22 |
| fact_aircraft_uplifts | 7,800 | 7,729 | 49 | 22 |
| fact_lpg_inventory_snapshot | 7,200 | 7,126 | 60 | 14 |
| fact_contract_pricing | 7,200 | 7,123 | 62 | 15 |
| fact_cashflow | 7,200 | 7,125 | 61 | 14 |
| fact_asset_downtime | 7,200 | 7,117 | 69 | 14 |
| fact_inventory_adjustments | 7,200 | 7,110 | 74 | 16 |
| fact_lubricant_inventory_snapshot | 6,600 | 6,513 | 75 | 12 |
| dim_customer | 6,000 | 5,852 | 148 | 0 |
| bridge_customer_hierarchy | 6,000 | 300 | 0 | 5,700 |
| fact_ev_charging_sessions | 6,000 | 5,886 | 102 | 12 |
| fact_working_capital | 5,400 | 5,337 | 44 | 19 |
| fact_vehicle_safety_events | 5,400 | 5,352 | 34 | 14 |

| Table | Raw rows | Cleansed | Quarantined | Duplicates removed |
|---|---|---|---|---|
| fact_contract_volume_commitments | 5,400 | 5,337 | 50 | 13 |
| fact_aviation_sales | 5,000 | 4,897 | 88 | 15 |
| dim_asset | 5,000 | 5,000 | 0 | 0 |
| bridge_asset_site | 5,000 | 5,000 | 0 | 0 |
| bridge_employee_site | 4,256 | 3,000 | 0 | 1,256 |
| fact_customer_service_cases | 4,200 | 4,154 | 38 | 8 |
| fact_asset_failures | 4,200 | 4,142 | 46 | 12 |
| fact_lpg_bulk_deliveries | 4,200 | 4,155 | 33 | 12 |
| fact_marine_sales | 4,000 | 3,921 | 71 | 8 |
| dim_pump | 4,000 | 4,000 | 0 | 0 |
| fact_maintenance_work_orders | 4,000 | 3,958 | 38 | 4 |
| fact_stock_losses | 3,600 | 3,550 | 35 | 15 |
| fact_shop_returns | 3,600 | 3,548 | 41 | 11 |
| fact_bunker_deliveries | 3,600 | 3,571 | 18 | 11 |
| bridge_site_product | 3,536 | 600 | 0 | 2,936 |
| dim_contract | 3,500 | 3,500 | 0 | 0 |
| bridge_customer_site | 3,172 | 1,500 | 0 | 1,672 |
| dim_employee | 3,000 | 3,000 | 0 | 0 |
| fact_near_misses | 2,760 | 2,729 | 24 | 7 |
| dim_tank | 2,500 | 2,500 | 0 | 0 |
| fact_capex | 2,400 | 2,369 | 26 | 5 |
| fact_promotions | 2,400 | 397 | 223 | 1,780 |
| fact_aviation_inventory_snapshot | 2,400 | 2,365 | 27 | 8 |
| bridge_route_site | 1,869 | 600 | 0 | 1,269 |

| Table | Raw rows | Cleansed | Quarantined | Duplicates removed |
|---|---|---|---|---|
| fact_marine_inventory_snapshot | 1,800 | 1,776 | 16 | 8 |
| fact_environmental_events | 1,320 | 1,296 | 18 | 6 |
| dim_equipment | 1,250 | 1,250 | 0 | 0 |
| bridge_supplier_product | 1,206 | 400 | 0 | 806 |
| dim_driver | 1,100 | 1,100 | 0 | 0 |
| bridge_site_promotion | 1,018 | 600 | 0 | 418 |
| dim_vehicle | 900 | 900 | 0 | 0 |
| fact_hsseq_incidents | 800 | 795 | 4 | 1 |
| bridge_site_hierarchy | 600 | 586 | 14 | 0 |
| dim_route | 600 | 600 | 0 | 0 |
| dim_site | 600 | 586 | 14 | 0 |
| fact_import_cargoes | 540 | 538 | 2 | 0 |
| dim_ev_charger | 400 | 400 | 0 | 0 |
| dim_supplier | 400 | 385 | 15 | 0 |
| dim_solar_asset | 300 | 300 | 0 | 0 |
| dim_city | 88 | 88 | 0 | 0 |
| dim_terminal | 60 | 60 | 0 | 0 |
| dim_promotion | 44 | 44 | 0 | 0 |
| dim_depot | 40 | 40 | 0 | 0 |
| dim_campaign | 38 | 38 | 0 | 0 |
| dim_product | 31 | 31 | 0 | 0 |
| bridge_product_hierarchy | 31 | 31 | 0 | 0 |
| dim_price_zone | 30 | 30 | 0 | 0 |
| dim_country | 29 | 28 | 1 | 0 |

| Table | Raw rows | Cleansed | Quarantined | Duplicates removed |
|---|---|---|---|---|
| dim_currency | 25 | 25 | 0 | 0 |
| dim_carrier | 22 | 22 | 0 | 0 |
| dim_warehouse | 20 | 20 | 0 | 0 |
| dim_airport | 12 | 12 | 0 | 0 |
| dim_legal_entity | 12 | 12 | 0 | 0 |
| dim_province | 9 | 9 | 0 | 0 |
| dim_port | 8 | 8 | 0 | 0 |
| dim_lpg_cylinder_type | 4 | 4 | 0 | 0 |



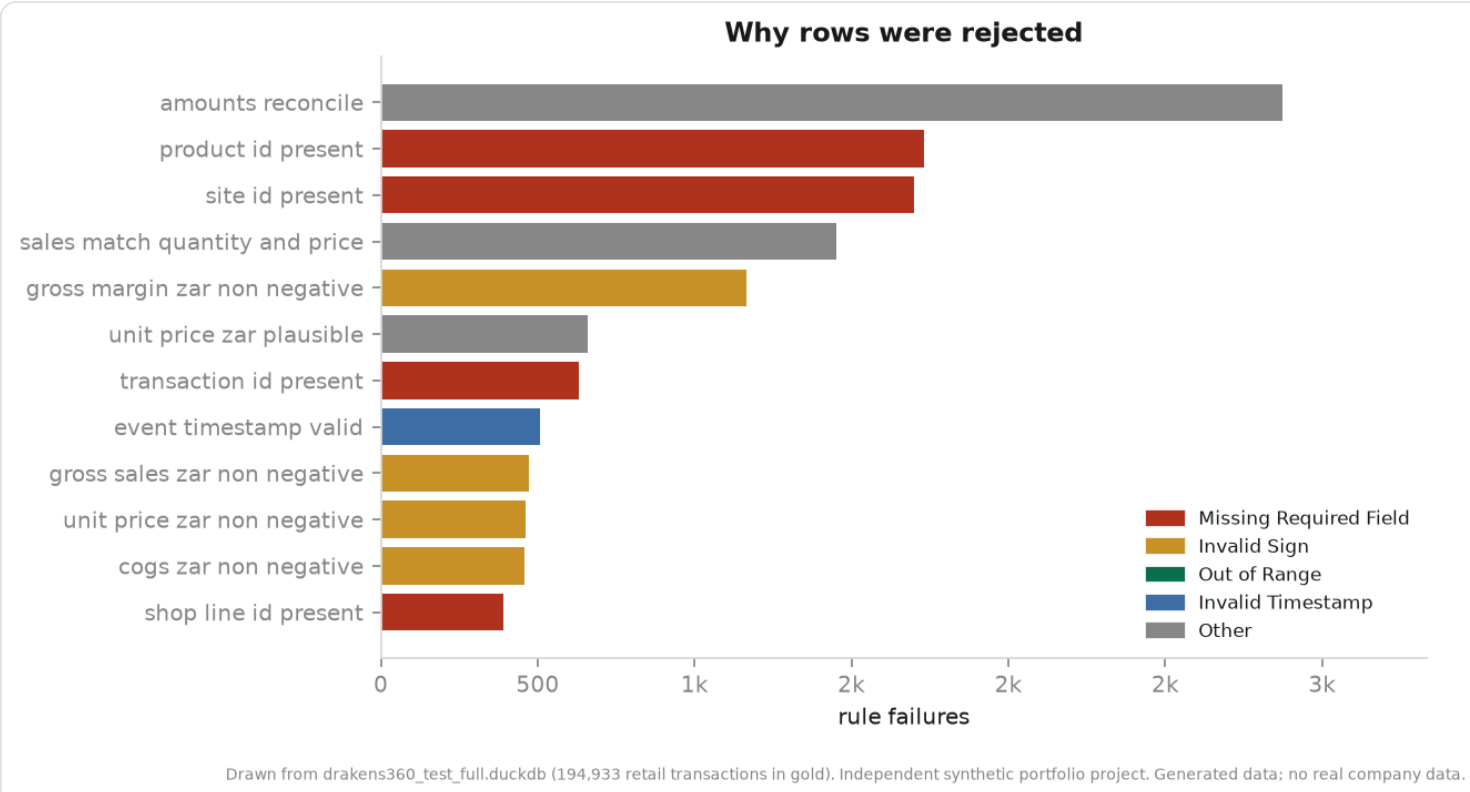

Every landed row is accounted for as cleansed, deduplicated or quarantined.

## Why rows were rejected

| Rule that rejected the row | Category | Likely owner | Rows |
| --- | --- | --- | --- |
| amounts_reconcile | Other | Data Platform | 2,873 |
| product_id_present | Missing Required Field | Source System | 1,731 |
| site_id_present | Missing Required Field | Source System | 1,699 |
| sales_match_quantity_and_price | Other | Data Platform | 1,452 |
| gross_margin_zar_non_negative | Invalid Sign | Business Process | 1,166 |
| unit_price_zar_plausible | Other | Data Platform | 660.00 |
| transaction_id_present | Missing Required Field | Source System | 630.00 |
| event_timestamp_valid | Invalid Timestamp | Source System | 506.00 |
| gross_sales_zar_non_negative | Invalid Sign | Business Process | 473.00 |
| unit_price_zar_non_negative | Invalid Sign | Business Process | 460.00 |
| cogs_zar_non_negative | Invalid Sign | Business Process | 458.00 |
| shop_line_id_present | Missing Required Field | Source System | 391.00 |
| journal_line_id_present | Missing Required Field | Source System | 311.00 |
| loyalty_transaction_id_present | Missing Required Field | Source System | 284.00 |
| litres_non_negative | Invalid Sign | Business Process | 253.00 |
| digital_event_id_present | Missing Required Field | Source System | 197.00 |
| reading_id_present | Missing Required Field | Source System | 176.00 |
| litres_within_range | Out of Range | Data Entry | 161.00 |
| snapshot_id_present | Missing Required Field | Source System | 160.00 |
| fill_pct_in_range | Other | Data Platform | 143.00 |
| tank_dip_id_present | Missing Required Field | Source System | 125.00 |
| points_value_zar_non_negative | Invalid Sign | Business Process | 117.00 |
| debit_amount_zar_non_negative | Invalid Sign | Business Process | 93.00 |
| order_id_present | Missing Required Field | Source System | 91.00 |

| Rule that rejected the row | Category | Likely owner | Rows |
| --- | --- | --- | --- |
| solar_reading_id_present | Missing Required Field | Source System | 91.00 |
| qualifying_spend_zar_non_negative | Invalid Sign | Business Process | 90.00 |
| customer_id_present | Missing Required Field | Source System | 88.00 |
| volume_litres_non_negative | Invalid Sign | Business Process | 88.00 |
| credit_amount_zar_non_negative | Invalid Sign | Business Process | 88.00 |
| capacity_litres_non_negative | Invalid Sign | Business Process | 83.00 |
| stock_on_hand_litres_non_negative | Invalid Sign | Business Process | 69.00 |
| volume_litres_within_range | Out of Range | Data Entry | 59.00 |
| delivery_id_present | Missing Required Field | Source System | 56.00 |
| revenue_zar_non_negative | Invalid Sign | Business Process | 43.00 |
| energy_kwh_non_negative | Invalid Sign | Business Process | 39.00 |
| ev_session_id_present | Missing Required Field | Source System | 33.00 |
| energy_kwh_within_range | Out of Range | Data Entry | 32.00 |
| fill_rate_pct_in_range | Other | Data Platform | 28.00 |
| list_price_zar_non_negative | Invalid Sign | Business Process | 27.00 |
| delivered_litres_non_negative | Invalid Sign | Business Process | 19.00 |
| delivered_litres_within_range | Out of Range | Data Entry | 13.00 |
| grid_cost_zar_non_negative | Invalid Sign | Business Process | 8.00 |
| work_order_id_present | Missing Required Field | Source System | 8.00 |
| total_cost_zar_non_negative | Invalid Sign | Business Process | 7.00 |
| rated_power_kw_non_negative | Invalid Sign | Business Process | 7.00 |
| labour_cost_zar_non_negative | Invalid Sign | Business Process | 7.00 |
| parts_cost_zar_non_negative | Invalid Sign | Business Process | 3.00 |
| average_power_kw_non_negative | Invalid Sign | Business Process | 3.00 |

| Rule that rejected the row | Category | Likely owner | Rows |
|---|---|---|---|
| incident_id_present | Missing Required Field | Source System | 2.00 |
| estimated_cost_zar_non_negative | Invalid Sign | Business Process | 1.00 |

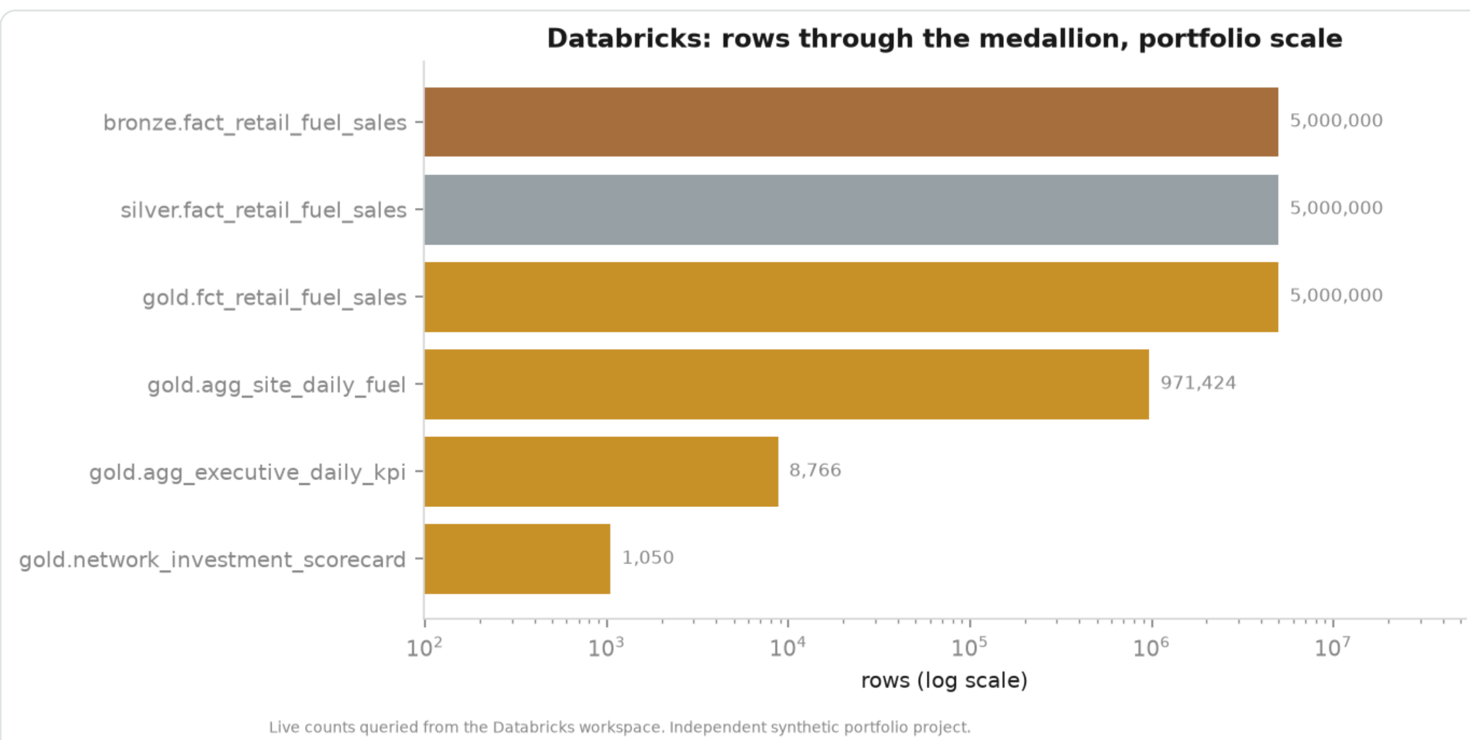



Rejection reasons. The operational response differs by rule.

## Warehouse

| Control | What it checks | Owner | Variance % | Result |
|---|---|---|---|---|
| DELIVERIES_HAVE_KNOWN_DESTINATION | Every delivery resolves to a site in the conformed dimension | Supply | 0.00 | PASS |
| GL_DEBITS_EQUAL_CREDITS | Posted journal debits equal credits across the ledger | Finance | 0.00 | PASS |
| ORDERS_HAVE_KNOWN_CUSTOMER | Every commercial order resolves to a customer in the dimension | Commercial | 0.00 | PASS |
| RETAIL_MARGIN_RECONCILES | Retail fuel gross margin equals revenue less cost of sales | Finance | 0.00 | PASS |
| RETAIL_VOLUME_TIES_TO_DAILY_AGG | Transaction-level litres tie to the site-day aggregate | Supply | 0.00 | PASS |

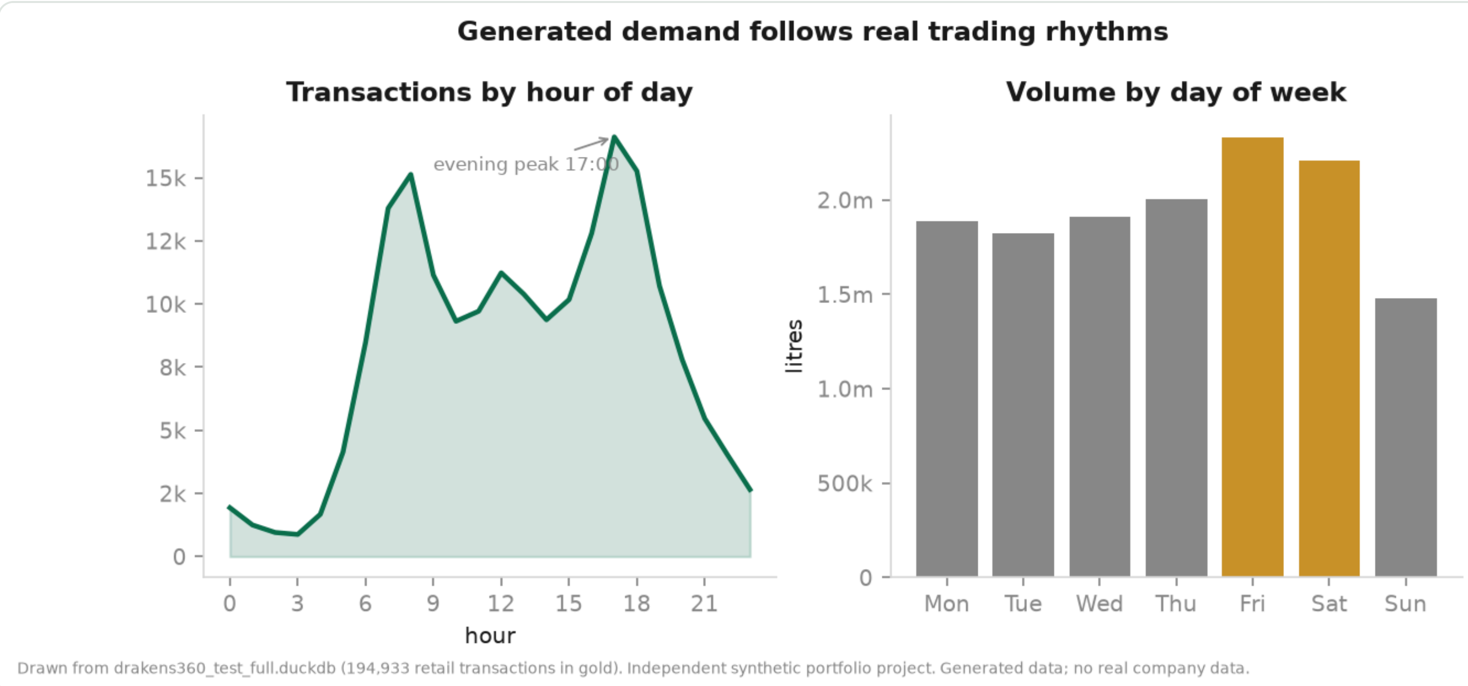
A reconciliation control evaluates a published number as data rather than only as a test, so a control that fails names the figure that is wrong and by how much. Controls that fail are shown as failures; a report that quietly omits them would defeat the purpose of having them.



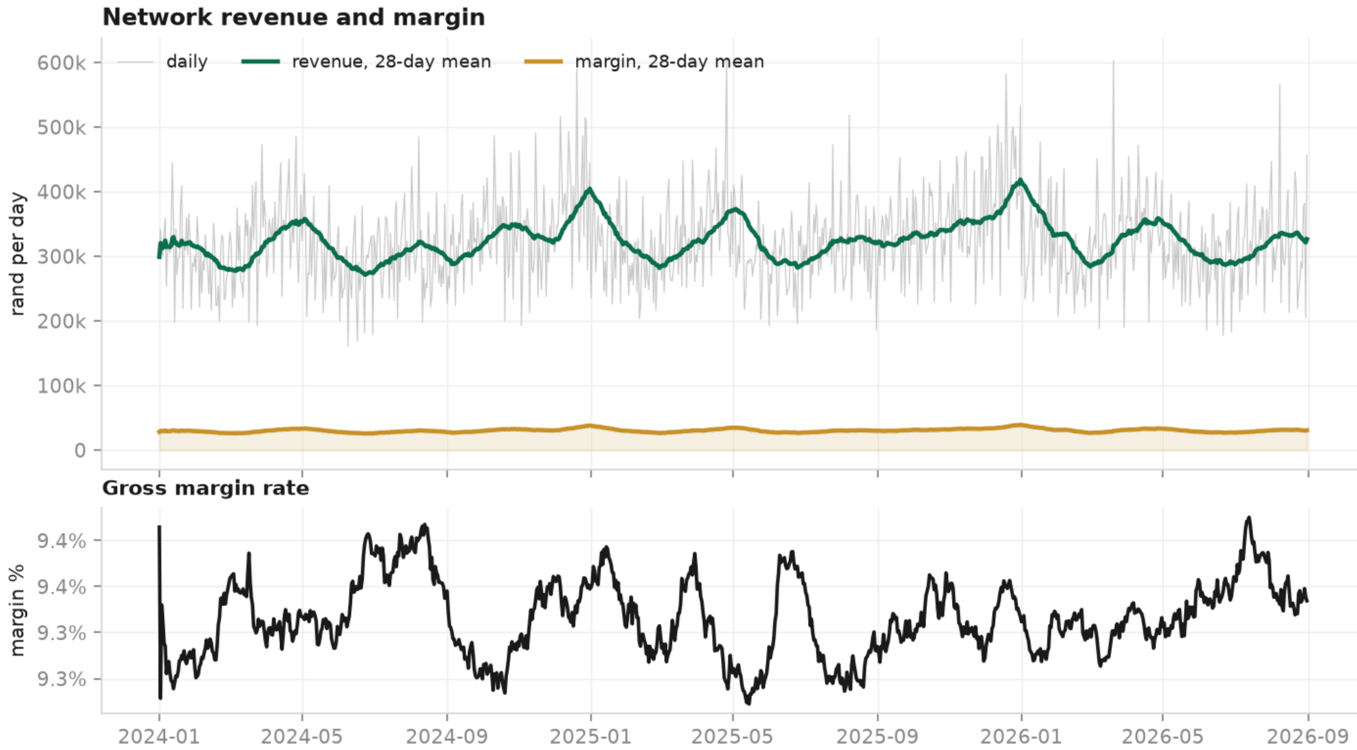

Live row counts through bronze, silver and gold on Databricks at portfolio scale.

# Demand and trading rhythm

A forecasting model can only learn a pattern the data actually contains. These three figures are the evidence that the generator produces one: an hourly commute shape, a weekly rhythm, and a southern-hemisphere seasonal cycle that survives all the way through to the warehouse.

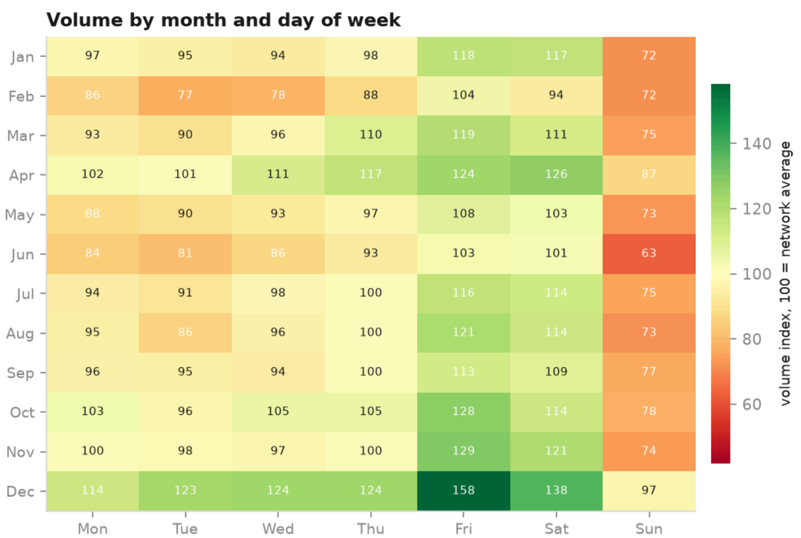



Commute peaks at 08:00 and 17:00, and a Friday-heavy week.




Drawn from drakens360_test_full.duckdb (194,933 retail transactions in gold). Independent synthetic portfolio project. Generated data; no real company data.

Revenue and margin across the modelled window, with the margin rate on its own axis. A rate and a total share an axis only in charts that say nothing about either.

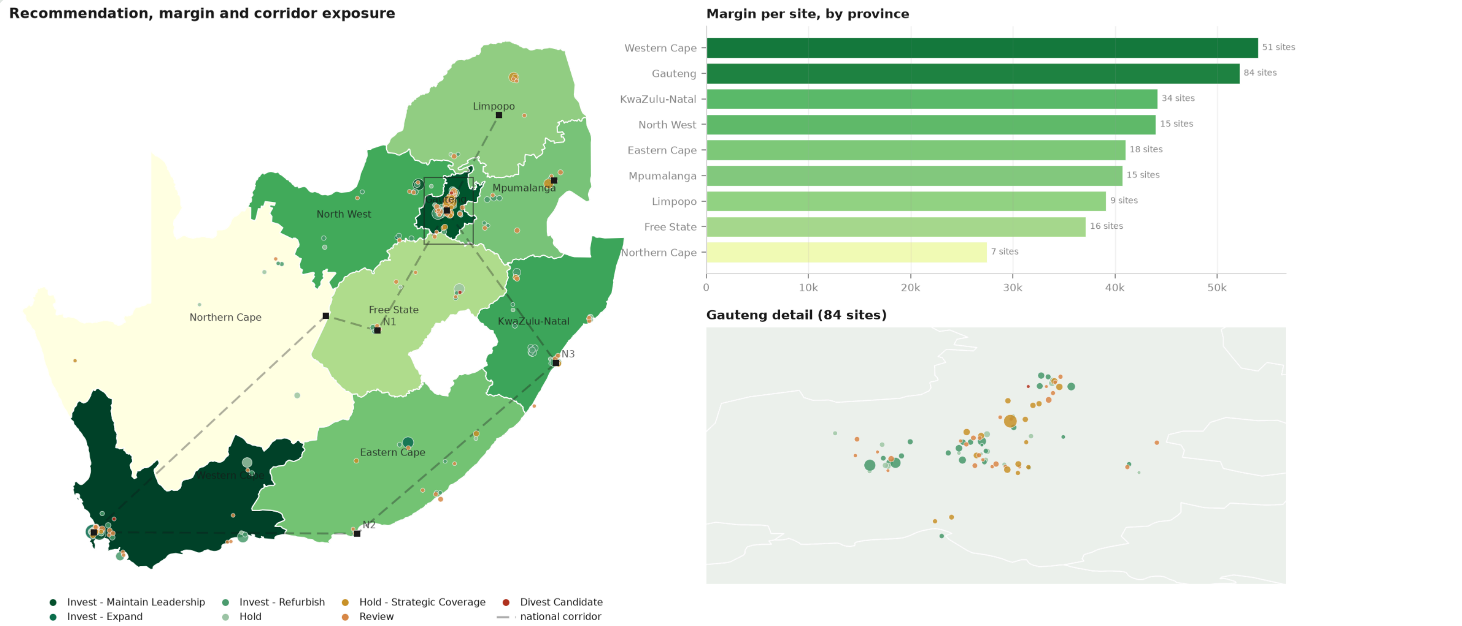



Mean litres per trading day, indexed to 100. Per day rather than per month because the modelled window ends in August: summed, September to December would read a third low and look like seasonality.

Drawn from drakens360_test_full.duckdb (194,933 retail transactions in gold). Independent synthetic portfolio project. Generated data; no real company data.

Month against day of week, indexed to the network average. December and January are the high-summer holiday peak; June and July the winter trough.

# Network and investment



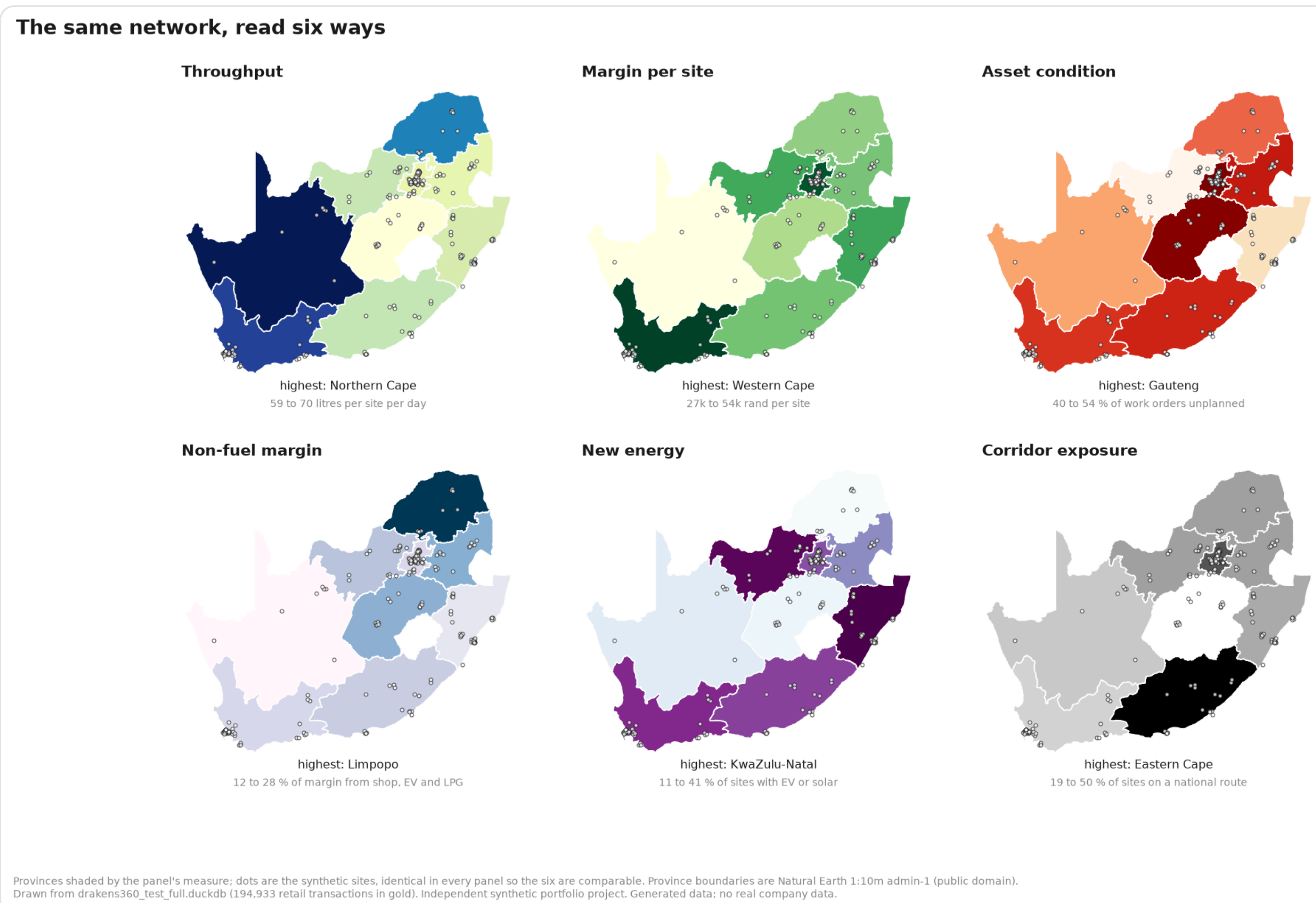

250 synthetic sites across 9 provinces -- the South African estate, 250 of 587 sites; the remaining 337 are in other African markets and have no province. Bubble area is total margin; colour is the investment recommendation. Province boundaries are Natural Earth 1:10m admin-1 (public domain); everything drawn on them is generated. Coordinates are South African town centroids plus random jitter and do not represent any real service station. Drawn from drakens360_test_full.duckdb (194,933 retail transactions in gold). Independent synthetic portfolio project. Generated data; no real company data.

Provinces shaded by margin per site; bubble area is total margin and colour is the investment recommendation. The N1, N2 and N3 are drawn because corridor sites are protected from divestment regardless of score. Province boundaries are Natural Earth 1:10m admin-1, public domain; everything drawn on them is generated.

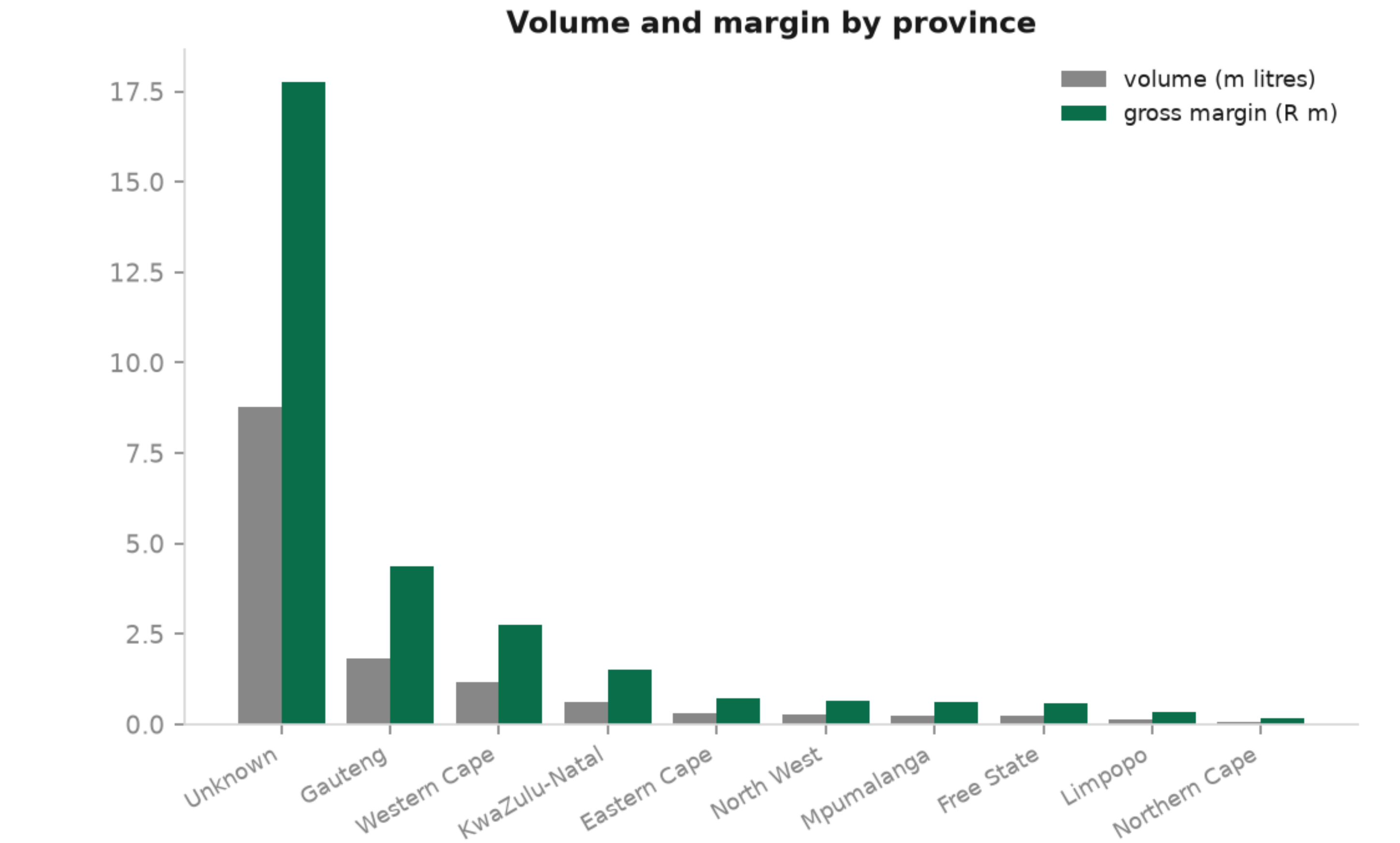
| Measure | Value |
|---|---|
| Budget | R1.50bn |
| Funded projects | 248 |
| Capital committed | R1.330bn |
| Expected annual uplift | R332.9m |
| Portfolio ROI | 25.0% per year |
| Blended payback | 4.0 years |
| Provinces covered | 9 of 9 |

## Plan by province

| Province | Projects | Capex (R m) | Uplift (R m) |
|---|---|---|---|
| Gauteng | 84 | 439.20 | 109.11 |
| Western Cape | 51 | 279.30 | 79.15 |
| KwaZulu-Natal | 34 | 188.00 | 44.93 |
| Eastern Cape | 16 | 87.30 | 21.48 |
| Free State | 16 | 85.60 | 19.52 |
| North West | 15 | 85.00 | 21.03 |
| Mpumalanga | 15 | 79.30 | 16.77 |
| Northern Cape | 7 | 42.20 | 11.74 |
| Limpopo | 9 | 38.20 | 8.19 |

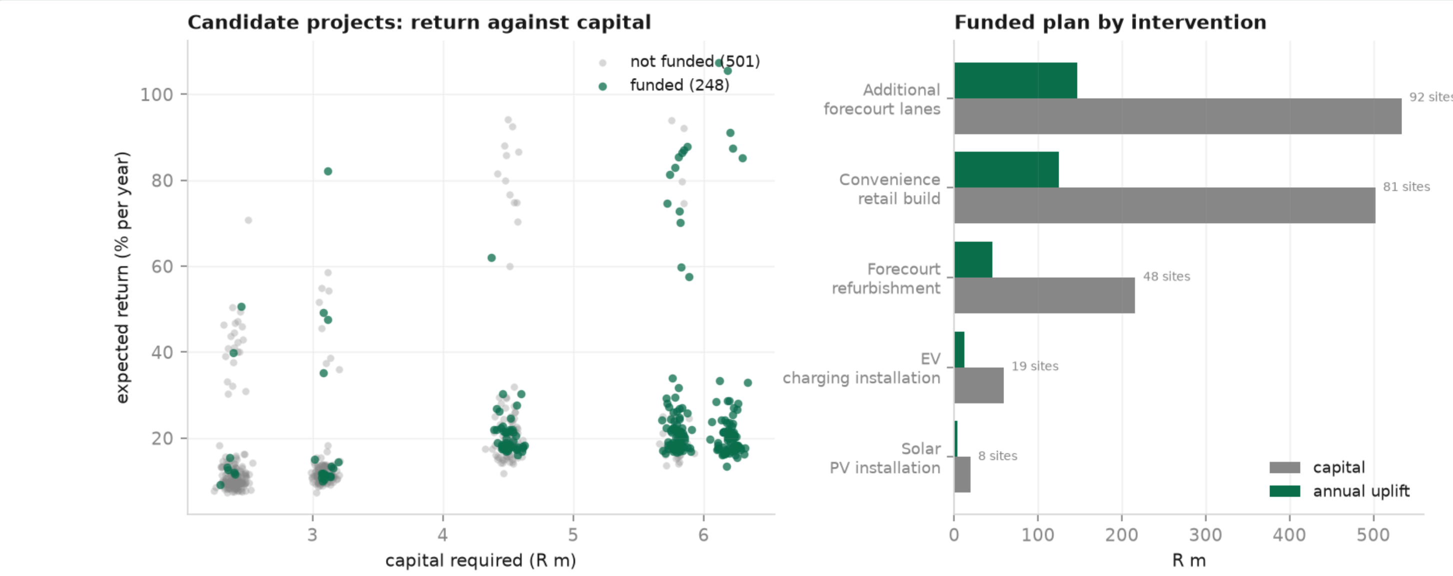



The same network read six ways, on a fixed basemap and extent so the panels are directly comparable. A single map answers one question; a network planner asks six.



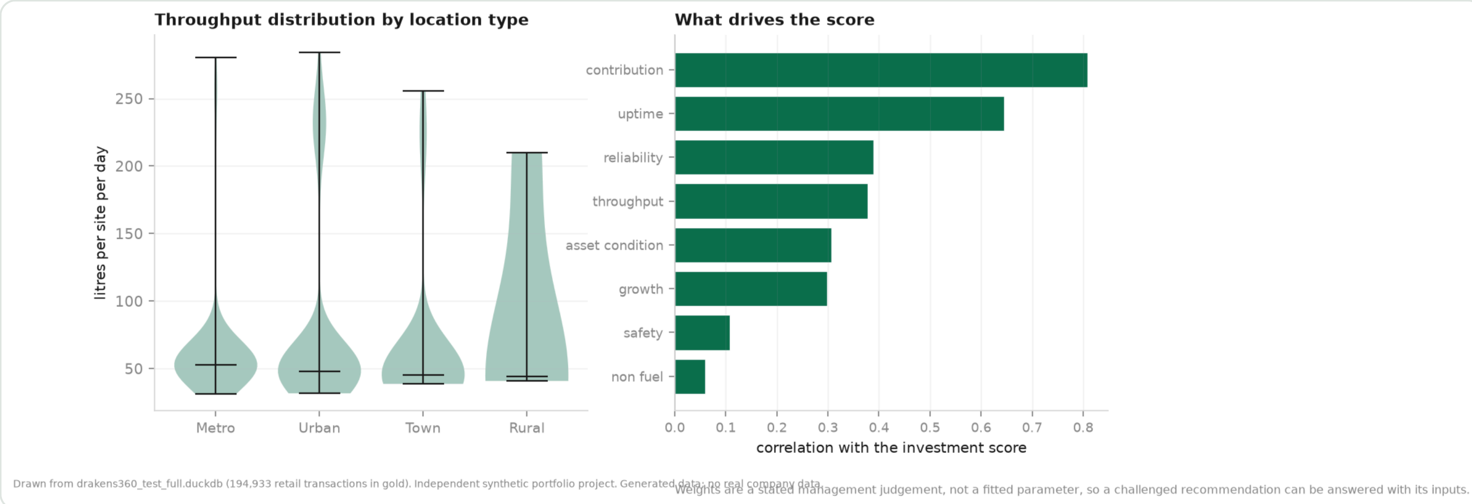

Drawn from drakens360_test_full.duckdb (194,933 retail transactions in gold). Independent synthetic portfolio project. Generated data; no real company data.

Volume and margin by province.




Drawn from drakens360_test_full.duckdb (194,933 retail transactions in gold). Independent synthetic portfolio project. Generated data; no real company data.

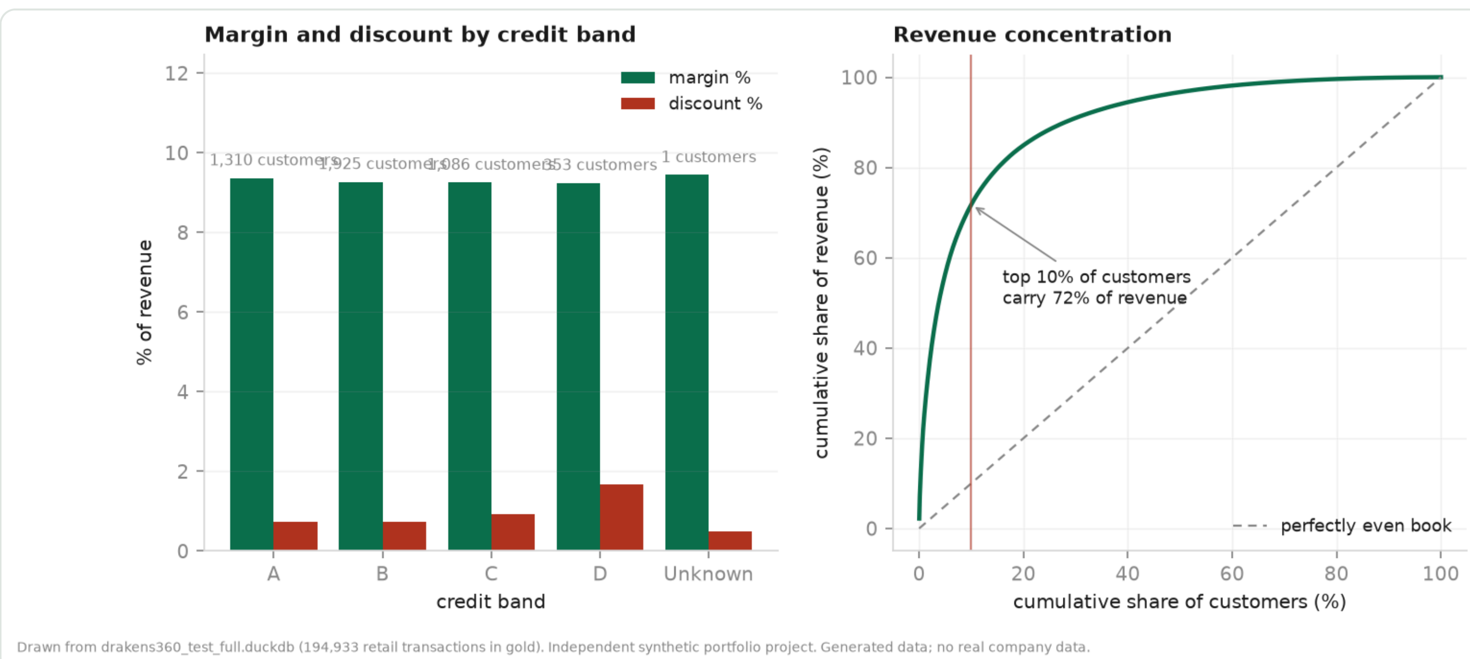
Every candidate project by return against capital, with the funded set highlighted. The funded set is not simply the top of the ROI ranking — provincial minimums and corridor protection pull specific projects in.



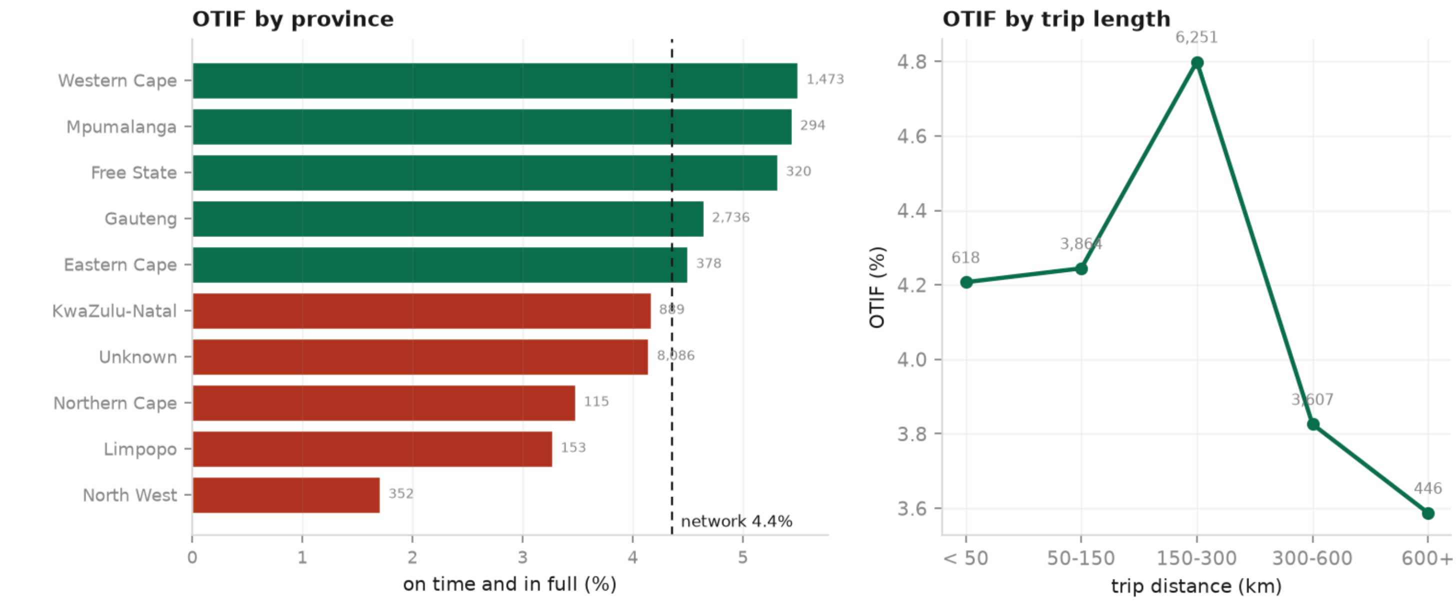

Left: throughput is skewed enough that a network average is a poor summary. Right: what actually drives the investment score, as the correlation between each component and the total.

# Commercial and operations

Four questions the business asks that the platform is built to answer, each computed once in gold rather than reassembled per report.

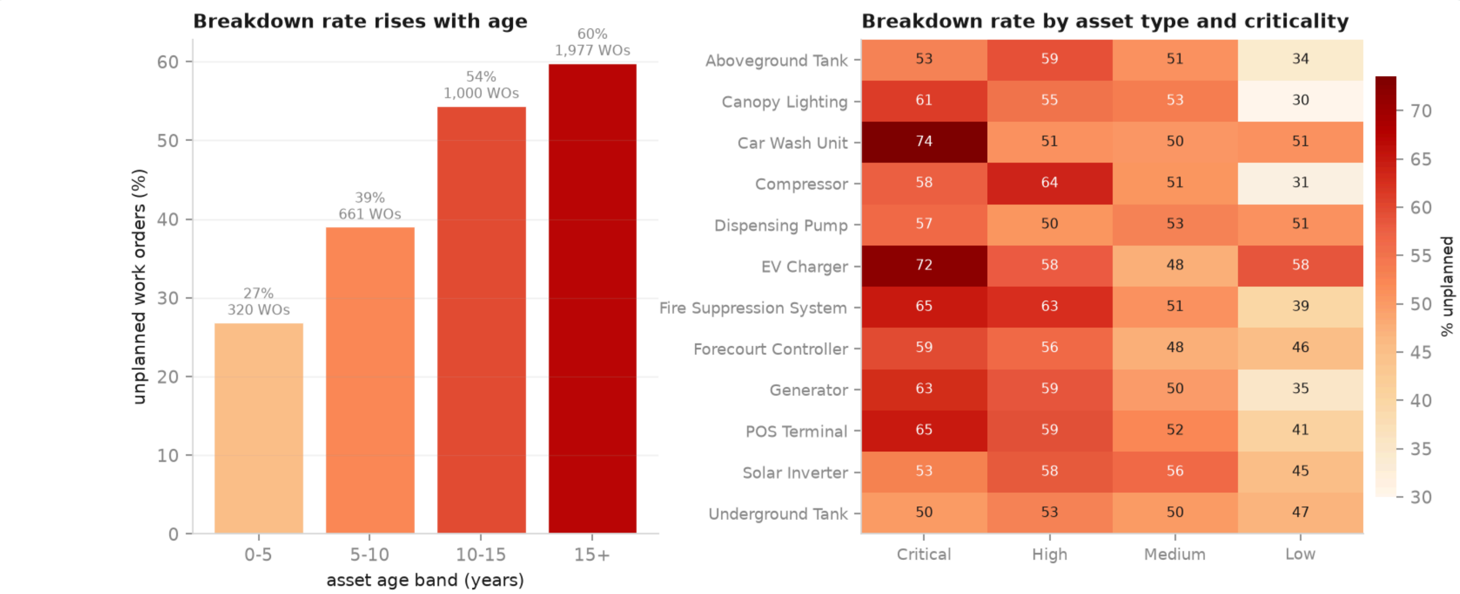



Left: whether the credit band means anything — worse-rated customers discount harder. Right: how concentrated the customer book is, which is a different risk from an even one.




Drawn from drakens360_test_full.duckdb (194,933 retail transactions in gold). Independent synthetic portfolio project. Generated data; no real company data.

OTIF broken out by province and by trip length. As a single network number it is the least useful form of it: broken out, it says whether lateness is a routing problem, a distance problem or one region's problem.



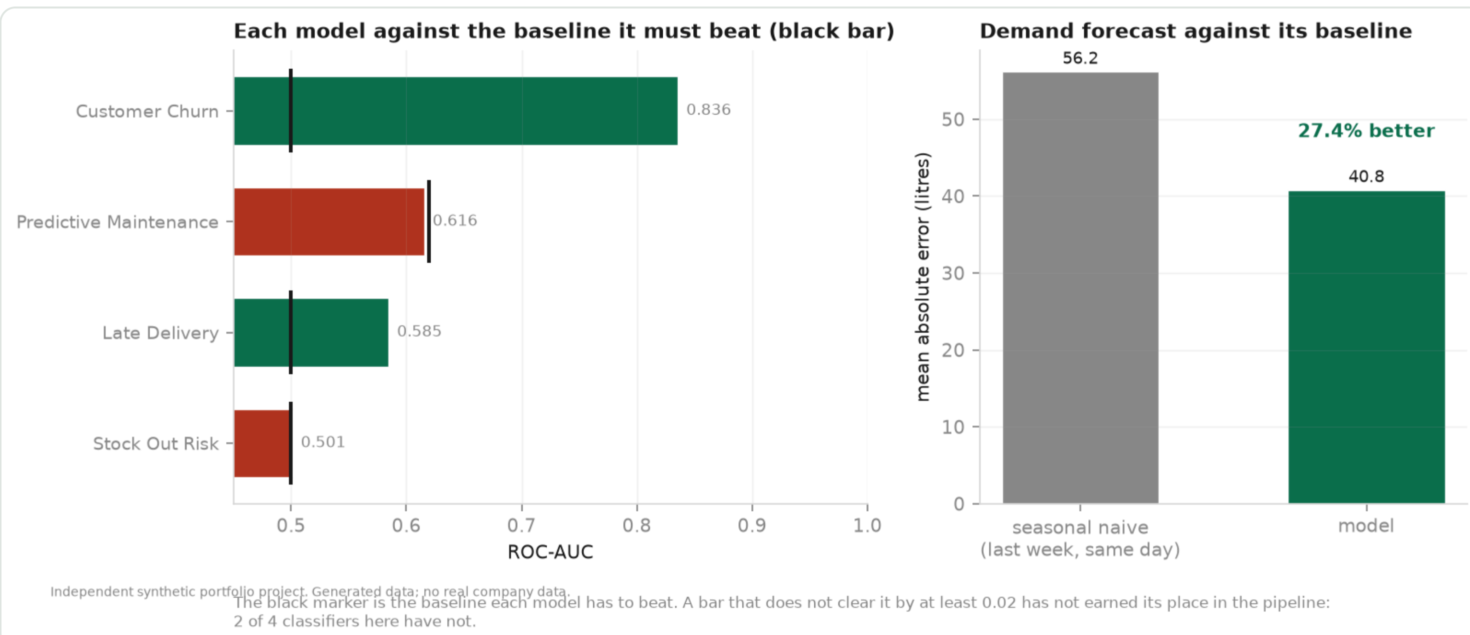

Drawn from drakens360_test_full.duckdb (194,933 retail transactions in gold). Independent synthetic portfolio project. Generated data; no real company data.

Breakdown rate against asset age and criticality. This is the signal the predictive-maintenance model is allowed to learn from, and the reason that experiment reports an age-only baseline.

## Machine learning

Six experiments, each reported against a baseline it has to beat. Not all of them win, and all six are listed — a portfolio that shows only the models that worked is not showing the job.

| Model | Headline result | Verdict |
|---|---|---|
| Customer Churn | ROC-AUC 0.836, 82% precision in the top decile | **Usable — ranks well above chance** |
| Demand Forecast | MAE 40.8 L against a seasonal-naive 56.2 L | **Beats its baseline by 27.4%** |
| Fraud Anomaly | 1.0% of shifts flagged; cash share 30.9% against 23.9% for normal shifts | Flagged shifts differ from normal ones on both axes |
| Late Delivery | ROC-AUC 0.585, 1.72x lift in the top decile | **Usable — the reviewed decile is enriched** |
| Predictive Maintenance | ROC-AUC 0.616, 1.19x lift in the top decile, against a baseline of 0.620 | **Does NOT beat its baseline** |
| Stock Out Risk | ROC-AUC 0.501 | **No usable signal** |




Each classifier against the baseline it has to beat, and the demand forecast against its seasonal-naive baseline. A bar that does not clear its black marker has not earned its place in the pipeline.

## What the platform found in itself

Each of these was caught by the project’s own tests, observability or baselines, which is the point of having them.

### A macro that silently corrupted every ratio

safe_divide('a - b', 'c') compiled to a - (b / c) because the macro did not parenthesise its arguments. Not an error — a plausible-looking wrong number. Caught by a range test asserting that dim_product.base_margin_pct falls between -5 and 100.

### A general ledger that did not balance

Journal lines were posted independently, so debits exceeded credits by 46%. Caught by obs_reconciliation_controls, which evaluates double-entry as data rather than only as a test. Fixed by emitting journals as balanced pairs.

### A partitioning choice that looked obvious and was wrong

The retail fact was partitioned on date_key, because every query filters on a date range. The maintenance job measured the result: 974 files averaging 0.2 MB, which OPTIMIZE can never merge because it does not cross partition boundaries. Under liquid clustering on (date_key, site_id) the same 5 million rows occupy 2 files averaging 94 MB — a 487-fold reduction in file count for identical data.

### A leak that scored 0.998

The stock-out model was scored against the current inventory snapshot. Its target is days_of_cover below a threshold; days_of_cover is stock over usage; and both components were features. Excluding days_of_cover, which the code did with a comment explaining why, excluded nothing — the model divided one component by the other and reproduced the rule. Re-pointed at the next snapshot it scores 0.510.

### A model with no baseline, losing to one variable

Predictive maintenance reported a 1.35x lift and passed review. It had no baseline. Ranking the queue by asset age alone scores ROC-AUC 0.620 against the model's 0.598, and the reported lift moves between 1.03 and 1.35 across capacity settings with no trend. Every experiment now computes and logs the baseline it has to beat.

### Null foreign keys where an unknown member belonged

The landing zone contains referential drift by design — a site code in a fact that the dimension never received. Gold left-joined those facts and kept the null key, which drops the row from every inner join and from any total grouped by province: 5,070 retail transactions with real litres and real margin were invisible to regional reporting. They now resolve to a -1 unknown member, so nothing disappears from a total and the drift is countable.

### A rule inferred from a column name, four times over

The pattern recurred often enough to be worth naming. A validity rule written against what a column is *called* rather than what it *means* destroys real data, and does it silently while every model builds and every test passes. `signed_amount_zar >= 0` quarantined every credit leg of the general ledger. A 2023-2027 timestamp window nulled 89% of site opening dates. An 8-45 rand fuel price band rejected 36% of the shop feed, where the same column name carries the price of a pie. Requiring a site on every row quarantined 58% of digital events, which are app sessions and were never at a station. Bands are now keyed by table and column, attribute dates get a window wide enough for a business history, and optional foreign keys are measured rather than assumed.

### A tab that split a dimension in two

The maintenance heatmap showed thirteen asset types where the dimension has twelve. clean_text called trim(), which in both DuckDB and Spark removes spaces and nothing else, so a tab-padded value reached gold as a separate category from the same value without one. Both real, both counted separately, neither obviously wrong in a total.

### Observability that measured 8% of what it claimed

obs_cleansing_summary named fourteen tables against a landing zone of 167, and the fourteen were whichever tables someone had thought to add — the least useful sample available. It is now generated by the same pass that generates the cleansing models: 128 monitored tables, 2,049,277 raw rows reconciling exactly to cleansed plus quarantined plus deduplicated, and zero tables failing reconciliation.

### A validity rule that deleted half the general ledger

The contract applied a blanket rule to every amount column: anything named *_zar must be non-negative. signed_amount_zar is negative on the credit side by design, so the rule quarantined 41,886 of the ledger's 84,000 lines — every credit leg of every journal. What survived was a ledger of debits with almost no credits, which reported a 99.98% imbalance that looked like a generator bug and was a contract bug. A validity rule inferred from a column name is a guess, and a wrong guess deletes data silently instead of failing loudly. Signed measures are now matched by naming convention and excluded, so a new net_ or _variance column is covered the day it is added.

### Double-entry integrity is a property of the journal, not the line

Cleansing assesses rows one at a time, so when one leg of a balanced pair failed its contract the other leg survived alone. Each orphaned leg sat in the ledger as an unmatched debit or credit, and the ledger could not balance however correct the generator was — the imbalance was created by the cleansing step itself. The mart now publishes only journals whose lines still net to zero, and the orphans stay in quarantine beside the leg that was rejected. Debits and credits now agree exactly, to the cent.

**A sentinel that passed every non-negativity check**

The generator uses 0 as one of its numeric sentinels, and 0 satisfies every 'amount >= 0' rule. Stripping zero the way -999 is stripped is not an option, because zero litres and zero margin are legitimate readings. The contract now checks amounts against their own row instead: a transaction of 82 litres at R22.34 reporting R0.00 of sales is internally inconsistent whatever caused it.

## Limitations

Stated plainly, because a report that claims completeness is misleading.

- The Databricks results are at portfolio scale (5,000,000 retail rows through bronze, silver and gold). The DuckDB warehouse figures in this report are at test scale, and each figure is stamped with the warehouse it came from. The two should not be read as one run.
- Predictive maintenance does not beat an age-only baseline, and stock-out risk has no signal at the current snapshot cadence. Both limits are in the generated data rather than in the estimators, and both fixes are changes to the generator: condition monitoring for the first, denser inventory readings for the second.
- The investment plan's rand figures rest on one stated benchmark — what a mid-sized site earns in fuel gross margin in a year — because the transaction fact is a sample and capital costs are not. Rank order and every ratio between sites are unaffected; only the level is rebased.
- Row filters are correct but their query cost under concurrent executive load has not been measured.
- There is no data subject access or erasure workflow. The masking and pseudonym design supports one; the process itself is not built.

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